🌱 Sustainable Tomorrow

☀️ Light ≡

📋

Comprehensive documentation of our data sources, analytical methods, and technical implementation for the **Viz Con 2025** contest at **Analyticon** - Amazon's premier conference on Data Engineering, Science and Analytics.

🎯 Project Overview

This interactive web application was developed by **@ygal** for the **Viz Con 2025** contest at **Analyticon** - Amazon's premier conference on Data Engineering, Science and Analytics. The project demonstrates advanced data visualization techniques using real climate datasets from NASA, NOAA, and other leading scientific institutions.

🏆 Viz Con 2025 Competition Strategy

- **Comprehensive Data Integration:** Five major climate datasets - NASA GISTEMP Temperature (1880-2024), NOAA CO_2 Concentrations, NSIDC Sea Ice Extent, IRENA Renewable Energy Statistics, and IEA Critical Minerals projections
- **Professional Development:** Production-ready React application with modern tech stack (React 18, Styled Components, Framer Motion, Recharts)
- **[...]telling:** Progressive disclosure of complex climate relationships [...] visualization types

🤖 Sustainability Chatbot

- **Accessibility & Inclusivity Excellence:**
  - WCAG 2.1 AA compliant design with high contrast ratios
  - Dark/Light theme support for visual accessibility
  - Color-blind friendly palette with distinct visual patterns
  - Audio accessibility through integrated climate podcast
  - Simple language explanations for complex scientific concepts
  - Keyboard navigation support and screen reader compatibility
- **Advanced Technology Stack:**
  - Frontend: React 18, Styled Components, Framer Motion
  - Visualization: Recharts, custom D3.js components
  - Data Processing: Real-time filtering, sorting, and analysis
  - Deployment: AWS Amplify with automated CI/CD
  - AI Integration: Amazon Q for development assistance
  - Audio: Integrated podcast player with climate education content
- **Amazon Standards:** Built to showcase data engineering, visualization expertise, and commitment to accessibility

👨‍💻 Development & Credit

Lead Developer: **@ygal**
AI Assistant: Amazon Q (for code optimization and data validation)
Conference: Analyticon - Amazon-wide Conference on Data Engineering, Science and Analytics
Contest: Viz Con 2025 - "Visualizing a Sustainable Tomorrow"
Year: 2025

📊 Data Sources & Processing

🌡️ Primary Data Sources

Dataset	Source	Time Range	Records	Purpose
Global Temperature	NASA GISTEMP v4	1880-2024	145 years	Temperature anomaly analysis
CO_2 Concentrations	NOAA Mauna Loa + Ice Core	1880-2024	145 years	Greenhouse gas correlation
Sea Ice Extent	NSIDC Sea Ice Index	1979-2024	46 years	Polar amplification analysis
Renewable Energy	IRENA Global Statistics	2000-2024	25 years	Clean energy transition
Critical Minerals	IEA Critical Minerals Data Explorer	2020-2050	11 minerals × 3 scenarios	Supply chain analysis

📈 Dataset Statistics

5
Major Datasets Integrated

145
Years of Temperature Data (1880-2024)

424
Current CO_2 Level (ppm, 2024)

+1.28°C
2024 Temperature Anomaly (vs 1951-1980 baseline)

3,500%
Solar Capacity Growth (2010-2024)

13.1%
Arctic Sea Ice Decline/Decade

11
Critical Minerals Tracked

25
Years of Renewable Energy Data

🤖 AI Integration & Analysis

🧠 Machine Learning Components

- **Pattern Recognition:** Identifying trends and anomalies in climate data
- **Correlation Analysis:** Statistical relationship quantification
- **Predictive Modeling:** Future scenario generation based on current trends
- **Natural Language Generation:** Automated insight creation
- **Ensemble Methods:** Combining multiple models for robust predictions

🔮 Prediction Methodology

Our AI prediction system uses ensemble modeling with three distinct scenarios:

- **Current Trends:** Linear extrapolation of existing patterns
- **Paris Agreement:** Policy-adjusted projections based on commitments
- **Aggressive Action:** Rapid decarbonization scenario modeling

Each prediction includes confidence intervals and uncertainty quantification. Models are validated against historical data with cross-validation techniques.

📊 Statistical Methods

- **Correlation Analysis:** Pearson correlation coefficients with significance testing
- **Trend Analysis:** Linear regression with confidence intervals
- **Time Series Analysis:** Seasonal decomposition and trend extraction
- **Regional Analysis:** Comparative statistics across climate zones
- **Risk Assessment:** Probabilistic modeling of climate impacts

💬 AI Chatbot Architecture

The integrated AI chatbot provides real-time climate insights using a comprehensive AWS serverless architecture designed for scalability and reliability.

- **Amazon Bedrock (Nova Pro Model):** Advanced large language model for natural language understanding and generation with comprehensive climate knowledge
- **API Gateway:** RESTful endpoint (https://z1odcbqh74.execute-api.us-east-1.amazonaws.com/prod/chat) with rate limiting and CORS support
- **AWS Lambda:** Serverless compute function processing chat requests with automatic scaling and error handling
- **Amazon S3:** Secure storage for comprehensive data sources including NASA, IRENA, IEA, NSIDC, and NOAA datasets
- **IAM Roles:** Least-privilege security model ensuring secure access between services
- **Knowledge Base:** Curated climate science data from authoritative sources with real-time updates
- **Response Time:** 2-5 seconds average with 99.9% availability and auto-scaling capabilities

💻 Technical Implementation

🏗️ Architecture Overview

The application follows a modern full-stack architecture with clear separation of concerns between data processing, API services, and user interface components.

React.js

Frontend framework for interactive UI

Chart.js

Data visualization and charting

Python Flask

Backend API and data processing

NumPy & Pandas

Scientific computing and analysis

Styled Components

Component-based styling

Framer Motion

Smooth animations and transitions

AWS Amplify

Automated deployment and hosting platform

AWS Infrastructure

Cloud hosting and services

🔧 Key Features

- **Responsive Design:** Mobile-first approach with adaptive layouts
- **Interactive Controls:** Real-time filtering and data exploration
- **Performance Optimization:** Lazy loading and efficient rendering
- **Accessibility:** WCAG 2.1 AA compliance with screen reader support
- **Progressive Enhancement:** Graceful degradation for older browsers

🎨 Design Philosophy

🎯 User Experience Principles

- **Progressive Disclosure:** Complex information revealed gradually
- **Clear Hierarchy:** Visual organization guides user attention
- **Contextual Help:** Explanations provided where needed
- **Consistent Interaction:** Predictable UI patterns throughout
- **Performance First:** Fast loading and smooth interactions

🌈 Visual Design System

- **Color Coding:** Consistent colors for data categories (red=temperature, blue=ice, green=renewable)
- **Typography:** Clear hierarchy with readable fonts
- **Glassmorphism:** Modern aesthetic with depth and transparency
- **Micro-interactions:** Subtle animations enhance usability
- **Data Visualization:** Clear, accurate, and engaging charts

👨‍💻 Project Credits & Recognition

🏆 Lead Developer & Architect

@ygal

Senior Business Operations Manager, AWS

Lead architect and developer of this comprehensive climate data visualization platform. Responsible for all technical implementation, data integration, UI/UX design, and scientific accuracy validation.

🤖 AI Development Partner

Amazon Q: AI assistant providing code optimization, data validation, and development acceleration. Helped ensure scientific accuracy and professional code quality throughout the development process.

🏢 Competition Context

> **Contest:** Viz Con 2025 - "Visualizing a Sustainable Tomorrow"
>
> **Conference:** Analyticon - Amazon-wide Conference on Data Engineering, Science and Analytics
>
> **Year:** 2025
>
> **Objective:** Demonstrate advanced data visualization and engineering capabilities using real climate datasets

📚 References & Acknowledgments

🔗 Authentic Data Sources

- **NASA Goddard Institute for Space Studies (GISTEMP v4):** Global temperature anomaly data (1880-2024)
- **NOAA Global Monitoring Laboratory:** Mauna Loa CO_2 measurements and ice core data
- **National Snow and Ice Data Center (NSIDC):** Sea Ice Index and Arctic analysis

- **International Renewable Energy Agency (IRENA):** Global renewable energy statistics
- **International Energy Agency (IEA):** Critical Minerals Data Explorer with demand projections for clean energy technologies
- **Intergovernmental Panel on Climate Change (IPCC):** Assessment reports and methodology guidance

🙏 Special Recognition

This project represents the exceptional work of **@ygal**, who single-handedly architected and developed this comprehensive climate data visualization platform for **Viz Con 2025** at **Analyticon**. The project showcases advanced data engineering, scientific rigor, and professional software development skills that exemplify Amazon's standards of excellence.

Special thanks to the global scientific community for providing open access to critical climate datasets, and to Amazon for fostering innovation through conferences like Analyticon and contests like Viz Con 2025.

"Built with passion for climate science and data visualization excellence by @ygal and his best friend Q 🤖 "

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental

Data Sources

🌡️ **Temperature Data:**
NASA GISTEMP v4 (Goddard Institute)
⚡ **Renewable Energy:**
IRENA Global Energy Statistics
⛏️ **Critical Minerals:**
IEA Critical Minerals Data Explorer

conservation through compelling data narratives using real datasets.

🧊 **Sea Ice Data:**
NSIDC Sea Ice Index
🏭 **CO2 Data:**
NOAA Mauna Loa Observatory

Technology & AI

Frontend: React.js, D3.js, Chart.js, Framer Motion
Backend: Python Flask, NumPy, Pandas
AI Integration: Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation
AI Chatbot: Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM
Deployment: AWS Amplify with automated CI/CD pipeline

Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics
Challenge: "Visualizing a Sustainable Tomorrow"
Approach: Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data
Goal: Inspire climate action through compelling data storytelling
Features: Interactive visualizations, correlation analysis, predictive modeling, and accessibility-first design

Built with 💚 by @ygal and his best friend Amazon Q 🤖

Powered by AWS | NASA, IRENA, IEA, NSIDC & NOAA data